

PUBLIC SUBMISSION

As of: March 21, 2025
Received: March 15, 2025
Status: [REDACTED]
Tracking No. m8b-3lqu-d36h
Comments Due: March 15, 2025
Submission Type: API

Docket: NSF_FRDOC_0001
Recently Posted NSF Rules and Notices.

Comment On: NSF_FRDOC_0001-3479
Request for Information: Development of an Artificial Intelligence Action Plan

Document: NSF_FRDOC_0001-DRAFT-9562
Comment on FR Doc # 2025-02305

Submitter Information

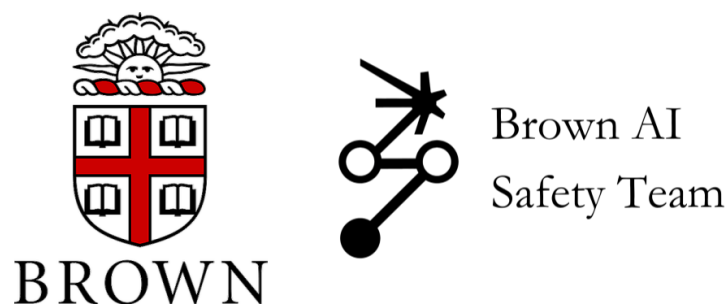
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General Comment

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Attachments

BAIST-OSTP-RFI-Comment



Response to the OSTP RFI: Advancing and Securing Biological Foundation Models

1 Executive Summary

Thank you for the opportunity to respond to the National Science Foundation’s and Office of Science and Technology Policy’s request for information regarding the “Development of an Artificial Intelligence Action Plan”.

The Brown AI Safety Team (BAIST) suggests that the current administration take decisive steps to both harness the transformative potential of biological foundation models for the U.S. bioeconomy and address associated security risks to ensure their responsible use. These models have the potential to accelerate innovation in biotechnology, synthetic biology, and biomanufacturing¹, positioning the U.S. as a global leader in these fields. However, realizing these benefits requires addressing biosecurity concerns that could otherwise hinder safe adoption and public trust. A proactive and rigorous assessment of security risks will be key to fostering a thriving bioeconomy while mitigating potential misuse.

We propose two key areas of action:

1. Evaluating Risks from Biological Foundation Models: The U.S. AI Safety Institute (USAISI) should lead an interagency initiative to comprehensively evaluate the risks of biological foundation models. These evaluations should determine the extent to which biological foundation models may lower barriers to bioweapon development and increase bioweapons capabilities. This includes assessing their role in enhancing pathogen design, evading medical countermeasures, and bypassing DNA sequence screening. Risk mitigation strategies such as data filtering and model unlearning should also be empirically tested. To ensure information security and biosafety and to facilitate public-private partnerships, the initiative should consist of multiple agencies such as DARPA, CDC, BARDA, NIH, and NIST.

2. Expanding the “Data for the Bioeconomy Initiative” to Include Biological Foundation Models: We recommend integrating biological foundation models into the existing Data Initiative under Executive Order 14081. This would involve establishing a structured framework for secure access, interoperability, and responsible use of these models within scientific and industrial applications. Establishing this infrastructure could allow the US bioeconomy to securely harness the full capabilities of biological foundation models.

Through these targeted actions, the U.S. can ensure that biological foundation models contribute to biotechnology and biomanufacturing advancements while addressing emerging security challenges. A proactive, evidence-based approach will allow the government to craft policies that enhance biosecurity

¹Wheeler. “Responsible AI in biotechnology: balancing discovery, innovation and biosecurity risks”. In: *Frontiers in Bioengineering and Biotechnology* 13 (2025), Article 1537471.

without unduly hindering technological progress.

2 Background on Biological Foundation Models

Biological foundation models are similar to large language models (LLMs), except they are trained on massive amounts of protein or DNA sequences instead of natural language. Biological foundation models have already revolutionized multiple scientific and industrial fields, such as protein engineering, synthetic biology, and genomics. For instance, AlphaFold² has achieved groundbreaking success in predicting protein structures with near-experimental accuracy, transforming structural biology and drug discovery. However, biological foundation models may simultaneously present unprecedented security challenges. Biological foundation models may both lower barriers to bioweapon development and increase bioweapon capabilities, thus creating unprecedented opportunities for malicious non-state actors³. As biological foundation models continue to rapidly scale up⁴, we anticipate huge opportunities for the US bioeconomy as well as increased security concerns.

3 Evaluating risks from Biological Foundation Models

3.1 Evaluation Framework

The misuse risks of biological foundation models have been discussed at length but lack comprehensive empirical evidence. To avoid regulation that would unduly stifle the US bioeconomy and instead promote evidence-based policies, we propose commissioning the US AI Safety Institute (USAISI) to evaluate to what extent biological foundation models can increase malicious misuse risks. These evaluations may focus on, but are not limited to, the following questions:

1. Given some fixed amount of resources (laboratory equipment and reagents, time, labor, etc), to what extent can a non-state actor enhance a pathogen’s virulence and transmissibility with the help of biological foundation models, compared to other AI and non-AI methods?
2. Conversely, how many fewer resources does a non-state actor need to surpass some target pathogen enhancement with the help of biological foundation models, compared to other AI and non-AI methods?
3. To what extent can a non-state actor use biological foundation models to digitally design functional pathogenic DNA sequences, without physical experiments, that do not get flagged by DNA sequence screening tools⁵ (see Section 3.2)?
4. To what extent can a non-state actor use biological foundation models to design pathogens that evade medical countermeasures, such as vaccines, rapid antigen tests, and monoclonal antibodies?

We also encourage evaluating how various safe model development approaches affect the above scenarios. Two notable safety approaches are:

1. **Filtering training data:** Remove DNA/protein sequences of concern from the model’s training data.
2. **Unlearning:** Implement unlearning methods⁶ after training that decrease the model’s usefulness in potentially harmful applications, such as designing enhanced viruses, while preserving neutral/beneficial applications, such as modeling drug-protein interactions.

²Jumper et al. “Highly accurate protein structure prediction with AlphaFold”. In: *Nature* 596 (2021), pp. 583–589.

³Wheeler, “Responsible AI in biotechnology: balancing discovery, innovation and biosecurity risks”.

⁴Maug et al. *Biological Sequence Models in the Context of the AI Directives*. 2024.

⁵Wittmann et al. “Toward AI-Resilient Screening of Nucleic Acid Synthesis Orders”. In: *bioRxiv* (2024).

⁶Zhang et al. “Unlearning Virus Knowledge Toward Safe and Responsible Mutation Effect Prediction”. In: *bioRxiv* (2024).

These approaches must be tested against fine-tuning procedures, such as fine-tuning on open-source mutation-effect datasets, as they may undo any safety gains.

We recognize that these evaluations may require or be strengthened by the physical synthesis of enhanced pathogens. To avoid accidental release of enhanced pathogens, we suggest a stratified approach:

1. **In Silico Evaluations:** Prioritize initial assessments and threat modeling in purely computational environments with appropriate cybersecurity measures.
2. **Harmless Biological Proxies:** Use harmless proxies to validate and refine computational models without risking accidental outbreaks. For instance, to evaluate threats of virus design, only use isolated proteins, pseudotyped viruses or virus-like particles, or viruses with low pathogenicity, such as adeno-associated viruses (AAVs). These experiments can be done in less secure labs, such as BSL-2 or BSL-3 labs⁷.
3. **Secure Pathogen Experiments:** Limit experiments using functional pathogens strictly to justified cases under stringent biosafety oversight.

We expect that a majority of evaluations can be done with the first and second approaches, albeit with some accuracy trade-offs.

We emphasize maintaining and strengthening the USAISI’s role rather than disbanding or diluting its responsibilities. However, we anticipate that the USAISI alone will not have the capacity or necessary expertise to execute physical experiments, ensure biosafety, and implement appropriate cybersecurity protocols. We thus encourage collaboration with other agencies, such as the Defense Advanced Research Projects Agency (DARPA), the Centers for Disease Control and Prevention (CDC), and the Biomedical Advanced Research and Development Authority (BARDA). We also suggest collaboration with the National Institutes of Health (NIH) and the National Institute of Standards and Technology (NIST) to facilitate public-private partnerships, particularly with academics in relevant areas and developers of biological foundation models.

3.2 DNA Sequence Screening

DNA sequence screening is an essential target for regulation, as successful bioweapon development by non-state actors is highly contingent on ordering DNA sequences from DNA synthesis companies. However, there are currently no widely-adopted standardized approaches to DNA sequence screening nor government mandates to implement effective screening. Securing sequence screening is becoming increasingly urgent as biological foundation models may enable non-state actors to entirely circumvent sequence screening. Red-teaming these screening tools by testing them on AI-designed sequences can reveal vulnerabilities and spur DNA synthesis companies to improve their protocols. Wittmann et al.⁸ carried out such an exercise. They worked directly with DNA synthesis companies to red-team their tools, revealed and reported a possible vulnerability to AI-designed sequences, and gave DNA synthesis companies enough time to fix their protocols before publishing the work. While their efforts are remarkable, we cannot solely rely on independent or industry-led initiatives to comprehensively secure sequence screening. Government agencies should work on these red-teaming experiments because they can operate on a much larger scale – they can convene many more DNA synthesis companies, carry out more involved experiments, and can more reliably manage infohazards.

3.3 Potential Policy Actions

These evaluations would provide high-quality evidence to guide (or discourage) policy. If these evaluations reveal substantial potential for the misuse of biological foundation models, we suggest exploring

⁷Centers for Disease Control, Prevention, and National Institutes of Health. *Biosafety in Microbiological and Biomedical Laboratories*. 2020.

⁸Wittmann et al., “Toward AI-Resilient Screening of Nucleic Acid Synthesis Orders”.

the following regulations and guidelines:

1. **Developer reporting requirements:** Mandate, incentivize, or suggest transparent documentation of model capabilities, training data sources, and safety protocols.
2. **Safety standards and benchmarks:** Establish clear criteria for acceptable levels of risk and necessary security features in biological foundation models.
3. **Access controls:** Develop tiered access systems to limit potential misuse without restricting legitimate scientific and industrial innovation. For instance, users in lower tiers can access attenuated models without capabilities related to enhancing pathogens, while users in higher tiers can access models with full capabilities. Access controls may be facilitated by creating centralized infrastructure that manages and integrates biological foundation models with an array of curated biological datasets, as discussed in Section 4.
4. **Improving oversight and safety of dual-use research:** Use results of safety evaluations to inform accurate risk assessments of legitimate research that uses biological foundation models to study pathogens. This could alleviate the oversight burden on federal funding agencies and researchers, following the United States Government Policy for Oversight of Dual Use Research of Concern and Pathogens with Enhanced Pandemic Potential. Furthermore, safety results can inform risk mitigation strategies for dual-use research. Specifically, biological foundation models can help replace physical experimentation with computational modeling to decrease the risk of accidental leakage.
5. **Improving DNA sequence screening:** As discussed in Section 3.2, safety evaluations of biological foundation models can reveal vulnerabilities in DNA sequence screening tools. Subsequent fixes can then inform amendments to government DNA sequence screening guidelines⁹. These fixes may include methods assisted by biological foundation models, which the USAISI-led interagency group may wish to research alongside safety evaluations.

In conclusion, carefully assessing the misuse risks of biological foundation models through structured evaluations and robust collaboration among public agencies, private entities, and academic institutions is crucial. Implementing transparent and informed regulatory measures, along with rigorous evaluations of safety approaches such as data filtering and model unlearning, can help mitigate potential threats. Additionally, proactively red-teaming DNA sequence screening protocols will enhance biosecurity preparedness and resilience. These coordinated actions will ensure the secure and responsible advancement of the bioeconomy without unnecessarily hindering growth and innovation.

4 Expanding the “Data for the Bioeconomy Initiative” to Include Biological Foundation Models

The “Data for the Bioeconomy Initiative” (Data Initiative), established under Section 4 of Executive Order 14081¹⁰, is a U.S. government-wide effort aimed at providing high-quality, accessible, interoperable, and secure biological data infrastructure. By collecting and curating diverse biological data sources, the Data Initiative aims to drive scientific discovery and innovation across health, agriculture, environmental science, and biomanufacturing.

We recommend expanding the scope of the Data Initiative to include biological foundation models. Similar to biological datasets, biological foundation models hold significant potential to support innovation by providing foundational capabilities for downstream applications in biotechnology, protein engineering, synthetic biology, and genomics.

⁹Office of Science and Technology Policy. *Framework for Nucleic Acid Synthesis Screening*. 2024.

¹⁰The White House. *Executive Order 14081: Advancing Biotechnology and Biomanufacturing Innovation for a Sustainable, Safe, and Secure American Bioeconomy*. 2022.

Packaging biological foundation models together with their associated datasets — both for initial training and fine-tuning — can substantially streamline research and accelerate scientific breakthroughs. For example, researchers in enzyme engineering could fine-tune a general protein model on enzyme function datasets. Integrating the resulting model into research and development could significantly reduce experimental time and resources.

Biological foundation models face similar challenges as biological datasets, including fragmentation across academic and industrial entities, substantial computational resource demands, and associated security risks. Applying FAIR (Findable, Accessible, Interoperable, and Reusable) principles can address these challenges:

- **Findable:** Develop a comprehensive registry of biological foundation models categorized by their intended applications, training data, safety practices, architectures, and computational requirements.
- **Accessible:** Provide basic computational resources for operating these models, considering institutional or cloud-based resources, possibly differentiating pricing structures for inference versus fine-tuning tasks.
- **Interoperable:** Standardize data formats for embedding and fine-tuning datasets, facilitating ease of use and integration across research contexts.
- **Reproducible:** Document training procedures, datasets, architectures, and safety evaluations, with appropriate precautions to manage information hazards.

Security is a significant concern for biological foundation models, given their potential misuse for bioweapons development by non-state actors (see Section 3 for more details). If safety evaluations reveal substantial risks, implementing tiered access controls could mitigate these risks:

1. Provide restricted, less capable model versions to lower-credential users, applying effective safety methods while granting full access to higher-credential users.
2. Clearly differentiate access between inference and fine-tuning, with stricter controls for fine-tuning.
3. Limit direct downloading of sensitive models; instead, use federated access, where users submit jobs to centralized, secure computational infrastructure. This environment can monitor inputs and outputs for subversive or unsafe actions.

Platforms like HuggingFace already facilitate access to biological foundation models. HuggingFace, however, lacks sufficient interoperability with a wide array of curated biological datasets. Integrating biological foundation models with future data infrastructure created through the Data Initiative would enable comprehensive interoperability. HuggingFace also does not emphasize security and safety. The Data Initiative could either guide or partner with HuggingFace to strengthen these areas or establish a dedicated platform inspired by precedents such as NIH’s dbGaP¹¹.

Public-private partnerships involving model developers, industry, and academia could further ensure practical, secure, and broadly useful implementations. These partnerships should define feasible approaches to balance open scientific innovation with necessary security measures, addressing both domestic and international considerations.

Ultimately, including biological foundation models within the Data Initiative or establishing a complementary initiative could significantly enhance U.S. leadership in biotechnology and biomanufacturing innovation while responsibly addressing the associated security challenges.

¹¹National Center for Biotechnology Information. *The database of Genotypes and Phenotypes (dbGaP)*.